

Shubham Bhattacharya

Software Engineer — Backend Systems (TypeScript, Node.js, PostgreSQL) · GenAI & RAG

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Available: Internship now · Full-time Dec 2026 · Open to relocate / remote

SUMMARY

Backend-focused software engineer with a GenAI/RAG specialization, working in **TypeScript (Strict)**, **Node.js**, **Express**, **Fastify**, and **PostgreSQL**. Designed, built, and deployed **3 solo personal projects**—an agentic contract analyser, a grounded RAG document service, and an autonomous support-agent backend—gated by **560+ automated tests** (318 + 249 across two repos) in GitHub Actions CI.

TECHNICAL SKILLS

Languages & Backend:	TypeScript (Strict), JavaScript (Node.js, Bun), Express 5, Fastify, Next.js, REST APIs, SSE Streaming
Databases & Caching:	PostgreSQL, pgvector, Redis (Upstash REST & ioredis), BullMQ Job Queues, Prisma ORM, SQL
GenAI & Retrieval:	Deterministic Quote Verification, Hybrid Search (BM25 + Dense Vectors), ReAct Agents, Tool Calling, Multi-Model Failover, Evals & Benchmarking, Langfuse Tracing
DevOps & Testing:	560+ Automated Tests (Unit, Integration) + Adversarial Golden-Set Evals, GitHub Actions CI/CD, Docker Compose, Sentry, Pino, Vercel, Render
AI-Assisted Dev:	Claude Code, Cursor (Autonomous Workflows, Spec-to-Code Generation)

FEATURED PROJECTS

ContractLens — Agentic Legal Contract Analyser | *Next.js 14, TypeScript (Strict), Express 5, Redis, BM25, Gemini, Groq, Langfuse, SSE*
Oct 2026 · Personal Project (Solo)

Live: agentic-contract-analyser-xbni.vercel.app | Code: github.com/shubhambhattacharya-dev/agentic-contract-analyser

- **Blocks displayed-citation hallucinations by construction** with a deterministic quote-verification engine that resolves every model answer to source byte offsets and page coordinates before render — unverified text cannot reach the UI, and every displayed quote is click-to-highlight auditable.
- **Accelerated contract research across up to 5 documents simultaneously** by designing an autonomous ReAct research agent loop with a **hard 4-round cap** and server-side trusted tool execution.
- **Automated financial risk detection in contract amendments**, creating clause-level comparison logic that flags high-risk numeric shifts (e.g., liability cap changing **\$100,000 → \$1,000,000 as CRITICAL**) regardless of identical phrasing.
- **Ship-gated by 318 automated tests (301 backend + 17 frontend)** running on every push in GitHub Actions (CI green as of Oct 2026); async **202 Accepted** ingestion with Redis-backed rate limiting absorbs burst traffic without dropping jobs.

RAG Chat — Grounded Document Intelligence | *Fastify, TypeScript, Bun, PostgreSQL, pgvector, Redis, BullMQ, SSE Streaming, Gemini, Groq*
Jul–Sep 2026 · Personal Project (Solo)

Live: ragchatbot-61jh.onrender.com | Code: github.com/shubhambhattacharya-dev/RagChatBot

- **Refuses out-of-scope queries instead of hallucinating** via strict distance-gated retrieval (≤ 0.5); **14/14 golden-set eval cases passed (Sep 3, 2026 committed run)**, including adversarial trap prompts that catch parametric-memory leakage — every eval run is committed as JSON in-repo, so results are dated and auditable rather than standing guarantees; **249 automated tests gate every merge**.
- **Offloaded 10–30s document parsing & embedding operations** from the main HTTP thread using an asynchronous BullMQ and Redis background queue worker.
- **Streams responses token-by-token over Server-Sent Events (SSE)**, paired with multi-provider waterfall failover (Groq → Gemini) so the service keeps working through provider outages and rate limits.

AI Ops — Autonomous Customer Support Backend | *Node.js, TypeScript, Express.js, PostgreSQL, Prisma ORM, Zod Validation, Groq, Sentry*
May–Aug 2026 · Personal Project (Solo)

Live: agentic-project-drab.vercel.app | Code: github.com/shubhambhattacharya-dev/AI-Ops

- **Automated multi-step customer support workflows** (order tracking, ticket escalation, refund drafts) using a custom ReAct agent loop built in pure TypeScript with zero heavy framework bloat.
- **Runtime Zod schema validation on every LLM argument** blocks malformed and hallucinated tool calls before any Prisma write lands — **0 invalid writes across the CI suite**.
- **Sustained agent availability through provider rate-limits and outages** with automated model failover routing (Llama 70B → 8B) and structured Pino + Sentry error telemetry.

EDUCATION

Bachelor of Computer Applications (BCA)

Vadodara, Gujarat, India

Parul University · Expected Dec 2026

CGPA: 8.13 (First Class with Distinction) · Relevant Coursework: Database Management Systems (DBMS), Operating Systems, Computer Networks, Data Structures & Algorithms, Software Engineering